Lecture 18 Scribe

L18.pdf
PDF

"Generate a lecture scribe suitable for exam preparation based ONLY on the provided context (Lecture 18 slides).Instructions:Use 'One-Shot Prompting'. I will provide you with an Example (The Shot) of how to convert a slide concept into a scribe format. You must follow this structure and style to generate the scribe for the Lecture 18 slides.EXAMPLE (THE SHOT):Input Context: 'Slide 1: Definition of Set. A set is a collection of distinct objects. Example: A = {1, 2, 3}. Notation: x ∈ A means x is in set A.'Output Scribe:'### 1. Fundamentals of SetsDefinition: A set is a well-defined collection of distinct objects.Notation:$A = \{1, 2, 3\}$ represents a set with elements 1, 2, and 3.$x \in A$ denotes that element x belongs to set A.'YOUR TASK:Now, generate the full lecture scribe for the attached Lecture 8 slides following the style and faithfulness of the example above.Faithfulness: Do not add outside information. Use exact definitions/notations from slides.Structure: Use clear headings (e.g., Sample Space, Events, Probability Axioms) as per the slides.Goal: Exam-ready reference material."

Below is the **exam-ready lecture scribe** generated strictly from the provided **Lecture 18 slides**. The structure and style follow the **one-shot example format** you provided, using headings, definitions, and formulas directly from the slides. All material is based only on the uploaded lecture file.

Lecture 18 Scribe

Marginal PMF, Correlation, Covariance, and Joint Gaussian Random Variables

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1. Marginal Probability Mass Function (Marginal PMF)

Definition

For discrete random variables X and Y, the **joint PMF** is defined as

$$p_{XY}(x, y) = P(X = x, Y = y)$$

The **marginal PMF** of a variable is obtained by **summing the joint PMF over all possible values of the other variable**.

$$p_X(x) = \sum_y p_{XY}(x, y)$$

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Key Idea:
The marginal PMF of one variable is obtained by **summing the joint PMF over all possible values of the other variable**.

Marginal PMF from a Joint PMF Table

Given a joint PMF table $p_{XY}(x, y)$:

- **Row sums give** $p_X(x)$
- **Column sums give** $p_Y(y)$

$$p_X(x_i) = \sum_j p_{ij}$$

$$p_Y(y_j) = \sum_i p_{ij}$$

Normalization conditions:

$$\sum_i p_X(x_i) = 1$$

$$\sum_j p_Y(y_j) = 1$$

Example

Joint PMF table:

$p_{XY}(x, y)$	$y = 0$	$y = 1$
$x = 0$	1/8	1/4
$x = 1$	3/8	1/4

Compute marginal PMFs

$$p_X(0) = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$$
$$p_X(1) = \frac{3}{8} + \frac{1}{4} = \frac{5}{8}$$
$$p_Y(0) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$$
$$p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Thus,

$$p_X(x) = \begin{cases} 3/8, & x = 0 \\ 5/8, & x = 1 \end{cases}$$

$$p_Y(y) = \begin{cases} 1/2, & y = 0 \\ 1/2, & y = 1 \end{cases}$$

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2. Correlation

Definition

A basic way to measure joint behavior between two random variables is the **product moment**.

$$R_{XY} = E[XY]$$

This represents the **average value of the product** XY and is often called a **raw correlation measure**.

Interpretation

- R_{XY} depends on the **scale of** X **and** Y.

- It is computed **about the origin**, not about the means.
- It mixes together:
  - mean values
  - spread
  - dependence

If

$$R_{XY} = 0$$

then X and Y are **orthogonal about the origin**.

To remove the effect of means, the variables are **centered**, which leads to **covariance**.

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3. Covariance

Definition

Covariance measures how two variables vary together after removing their means.

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

where

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Interpretation

Covariance indicates the **direction of joint linear variation** between variables.

Cases:

1. **Positive covariance**

$$\mathrm{Cov}(X, Y) > 0$$

Deviations from the means tend to have the **same sign**.

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2. **Negative covariance**

$$\mathrm{Cov}(X, Y) < 0$$

When one variable increases, the other tends to **decrease**.

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3. **Zero covariance**

$$\mathrm{Cov}(X, Y) = 0$$

There is **no linear co-variation** between variables.

Important note:

- Covariance indicates **direction** of relationship.
- The **magnitude depends on units**, so it cannot measure relationship strength directly.

Algebraic Form of Covariance

Start with

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Expanding:

$$= E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y]$$

$$= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y$$

Since

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Therefore

$$\mathrm{Cov}(X, Y) = E[XY] - \mu_X\mu_Y$$

or equivalently

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Properties of Covariance

1. **Symmetry**

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

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2. **Variance as a special case**

$$\text{Cov}(X, X) = \text{Var}(X)$$

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3. **Linearity with respect to constants**

$$\text{Cov}(aX + b, cY + d) = ac\,\text{Cov}(X, Y)$$

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4. **Independence implies zero covariance**

If

$$X \perp Y$$

then

$$\text{Cov}(X, Y) = 0$$

However,

zero covariance means uncorrelated but does not necessarily imply independence.

Example 1

If

$$Y = -2X + 3$$

Find $\text{Cov}(X, Y)$.

Using the covariance formula:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Since

$$XY = X(-2X + 3) = -2X^2 + 3X$$

$$E[Y] = E[-2X + 3] = -2E[X] + 3$$

Substituting into covariance formula:

$$\text{Cov}(X, Y) = -2E[X^2] + 2(E[X])^2$$

Using

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

We obtain

$$\text{Cov}(X, Y) = -2\text{Var}(X)$$

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4. Pearson's Correlation Coefficient

Pearson's coefficient measures the **strength and direction of the linear relationship** between two variables.

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

or

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Properties:

- Normalized version of covariance
- Unit-free
- Measures strength and direction of linear relationship

Range:

$$-1 \leqslant \rho_{XY} \leqslant 1$$

Interpretation:

- $\rho_{XY} > 0 \rightarrow$ positive linear trend
- $\rho_{XY} < 0 \rightarrow$ negative linear trend
- $\rho_{XY} = 0 \rightarrow$ no linear correlation

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5. Example 2 (Joint PDF)

Given the joint PDF

$$f_{XY}(x,y) = \begin{cases} \frac{xy}{96}, & 0 < x < 4,\ 1 < y < 5 \\ 0, & \text{otherwise} \end{cases}$$

Compute:

1. $E[X]$
2. $E[Y]$
3. $E[XY]$
4. $E[2X + 3Y]$
5. $Var(X)$
6. $Var(Y)$
7. $Cov(X, Y)$
8. ρ_{XY}

Results

$$E[X] = \frac{8}{3}$$

$$E[Y] = \frac{31}{9}$$

$$E[XY] = \frac{248}{27}$$

$$E[2X + 3Y] = \frac{47}{3}$$

$$Var(X) = \frac{8}{9}$$

$$Var(Y) = \frac{92}{81}$$

$$Cov(X, Y) = 0$$

$$\rho_{XY} = 0$$

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6. Jointly Gaussian Random Variables

Definition

A pair (X, Y) is **jointly Gaussian** if:

- Its **joint density follows a bivariate Gaussian distribution**
- The **marginal distributions are Gaussian**.

Joint Gaussian PDF

For variables X and Y:

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)\right.\right.$$

Where

- $\mu_X, \mu_Y \rightarrow$ means
- $\sigma_X, \sigma_Y \rightarrow$ standard deviations
- $\rho \rightarrow$ Pearson correlation coefficient

Real-Life Interpretation

Example:

- X: Atmospheric CO_2 concentration (ppm)
- Y: Global surface temperature anomaly (°C)

Because both variables are influenced by many random environmental processes, their joint behavior can often be approximated by a **Joint Gaussian Distribution**.

If

$$\rho > 0$$

then **higher CO_2 levels tend to occur with higher temperatures**.

✅ **End of Lecture 18 Scribe**

If you want, I can also convert this into:

- **2-page ultra-compressed exam cheat sheet**, or

- **step-by-step solved formula sheet for exams**. These are extremely helpful for probability courses.